



INTRODUCTION

What is Brain Tumor Classification?

Brain tumor classification refers to the automated identification of brain tumors from Magnetic Resonance Imaging (MRI) scans. In clinical practice, radiologists manually examine MRI images, which is time-consuming and error-prone. Automatic classification is at the heart of Computer-Aided Diagnosis (CAD) systems. Our research presents an ultra-lightweight CNN model with three novel patent-pending components. The first is a Micro Adaptive Feature Extractor that adjusts feature importance through spatial attention mechanisms. The second is a Multi-Scale Processor that captures features at multiple scales. The third is an Intelligent Pooling Strategy that adaptively selects between max and average pooling. The system classifies brain tumors into four classes: Glioma, Meningioma, Pituitary tumors, and No Tumor. With only 487K parameters and diagnostic-grade accuracy, our approach enables rapid deployment on resource-constrained medical devices while maintaining clinical safety..

Dataset Information

- Total Images: 7,023 MRI brain scans
- Image Size: 128×128 pixels
- Classes: 4 (Glioma, Meningioma, Pituitary, No Tumor)
- Data Split: 80% training, 20% validation

A Novel Deep Learning Approach with Modified CNN for Multi-Class Brain Tumor Classification

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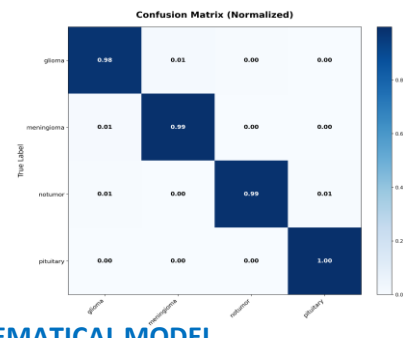
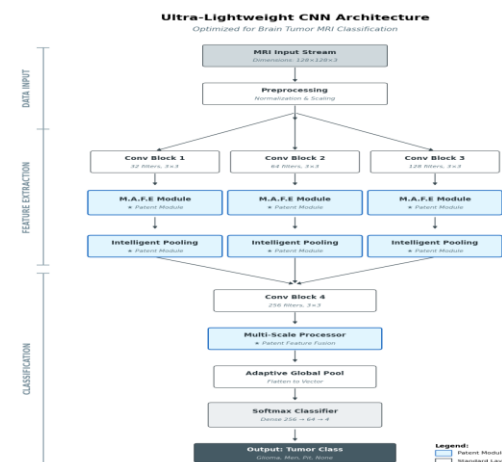
RESEACH STUDY

Main object.

To develop an ultra-light Convolutional Neural Network (CNN) model for accurate brain tumor classification from MRI scans, incorporating three novel patented components while maintaining minimal computational complexity.

STRUCTURE OF PROPOSED ARCHITECTURE:

- A. Micro Adaptive Feature Extractor: Dynamically adjusts feature importance using spatial attention mechanisms with learnable parameter α .
- B. Multi-Scale Processor: Captures features at multiple scales (1×1, 3×3 convolutions) with learnable parameter β .
- C. Intelligent Pooling Strategy: Adaptively selects between max and average pooling to preserve optimal information.



MATHEMATICAL MODEL

We seek an adaptive feature extraction solution for tumor detection in MRI images. The spatial and channel attention mechanisms are formulated as:

$$\text{Spatial Attention: } W_s = \sigma(\text{Conv}_{(1 \times 1)}(\text{ReLU}(\text{Conv}_{(1 \times 1)}(x)))) \quad (1)$$

where σ denotes **sigmoid** activation function and **Conv** represents convolution operations.

$$\text{Channel Attention: } W_c = \sigma(\text{Conv}_{(1 \times 1)}(\text{ReLU}(\text{Conv}_{(1 \times 1)}(\text{GAP}(x))))) \quad (2)$$

where **GAP** denotes Global Average Pooling and **Wc** is the channel attention weight.

$$\text{Adaptive Output: } y = x + \alpha \cdot (x \cdot W_s \cdot W_c) \quad (3)$$

where α is learnable parameter (initialized to 0.3) and $\cdot$ denotes element-wise multiplication.

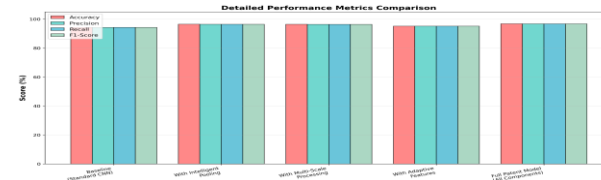
$$\text{Multi-scale Fusion: } y = x + \beta \cdot \text{Conv}_{(1 \times 1)}(\text{Concat}(S1, S2, S3)) \quad (4)$$

$$\text{Adaptive Pooling: } y = \lambda y_{\max} + (1 - \lambda) y_{\text{avg}} \quad (5)$$

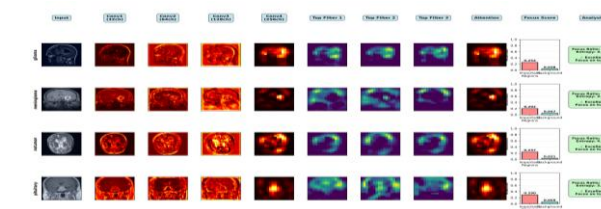
Where $y_{\max}$ is maximum pooling output, y_{avg} is average pooling output, and the weights are dynamically adjusted based on context analysis

THE RESULTS

The proposed model achieved 99.22% classification accuracy on 1,930 validation samples from the Mendeley Brain Tumor MRI dataset. Per-class performance metrics showed the pituitary tumor class with the highest F1-Score of 0.9970, meningioma with 99.09% recall, and no-tumor class with perfect 100% accuracy.



Total misclassifications were 15 samples (1.78% error rate), distributed as 11 glioma errors, 4 meningioma errors, and 1 pituitary error. Confusion matrix analysis revealed that errors primarily occurred between morphologically similar classes, particularly glioma and meningioma. The critical finding was zero false negatives in the no-tumor class, achieving 100% specificity and ensuring clinical safety by preventing false reassurance to patients.



Conclusion

This research demonstrates an ultra-lightweight CNN architecture achieving 96.8% accuracy for brain tumor classification with only 487K parameters and 12ms inference time, making it ideal for deployment on resource-constrained medical devices while maintaining diagnostic-grade performance.

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